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An interpretable polytomous cognitive diagnosis framework for predicting examinee performance

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ABSTRACT

As a fundamental task of intelligent education, deep learning-based cognitive diagnostic models (CDMs) have been introduced to effectively model dichotomous testing data. However, it remains a challenge to model the polytomous data within the deep-learning framework. This paper proposed a novel **P**olytomous **C**ognitive **D**iagnosis **F**ramework (PCDF), which employs **C**umulative **C**ategory **R**esponse **F**unction (CCRF) theory to partition and consolidate data, thereby enabling existing cognitive diagnostic models to seamlessly analyze graded response data. By combining the proposed PCDF with IRT, MIRT, NCDM, KaNCD, and ICDM, extensive experiments were complemented by data re-encoding techniques on the four real-world graded scoring datasets, along with baseline methods such as linear-split, one-vs-all, and random. The results suggest that when combined with existing CDMs, PCDF outperforms the baseline models in terms of prediction. Additionally, we showcase the interpretability of examinee ability and item parameters through the utilization of PCDF.

1. Introduction

In the digital era, the use of cognitive diagnosis as a crucial evaluation approach is gradually drawing attention (Liu, 2021). It serves to gauge individuals' cognitive states and abilities, offering robust tools applicable across diverse domains including education (Qi et al., 2023; Wang et al., 2020), psychology (Muraki, 1992; Templin & Henson, 2006), gaming (Shih, Kuo, & Lee, 2019), and clinical measurement (de la Torre, van der Ark, & Rossi, 2018; Liang et al., 2023). Within these domains, cognitive diagnosis not only helps to evaluate learning levels and psychological characteristics, but also plays an important role in guiding instructional strategies, assisting with diagnosis processes, and improving overall user experience. Testing is a common way to assess cognitive function. Test takers respond to a series of questions, and cognitive diagnostic models (CDMs) use their responses to estimate their potential abilities across various cognitive dimensions.

Polytomously scored items, as opposed to dichotomously scored items, can provide richer diagnostic information by delving deeper into specific problem-solving steps or examinee processing (Ma & de la Torre, 2016). In educational assessments, subjective problems and composition are typically used to grade scores to indicate different levels of mastery. Likewise, Likert-type items are commonly utilized in psychological tests to evaluate the diverse levels of examinees' psychological states. Taking into account the characteristics of polytomously scored data, researchers developed numerous cognitive diagnostic models based on

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statistical methods, including the Graded Response Model (GRM) (Samejima, 1969), the Multidimensional Graded-Response Model (MGRM) (de Ayala, 1994), the Generalized Partial Credit Model (GPCM) (Muraki, 1992), and the General Polytomous diagnosis model (GPDM) (Chen & de la Torre, 2018).

In recent years, numerous computer academics have focused on the advantages of CDMs in intelligent tutoring systems. This emerging focus presents a fresh perspective for advancing the analysis of educational testing data. Deep learning-based CDMs, leveraging the robust feature extraction and representation capabilities of deep learning technology, excel in extracting features from a vast array of sparse data points, capturing more complex feature interactions, and integrating multi-modal data. The NCDM (Neural Cognitive Diagnostic model) is the typical deep learning-based CDM that models the complex non-linear interactions between the exercise factor and the students' factor using multiple neural layers (Wang et al., 2020). Inspired by NCDM, numerous deep learning-based CDMs have been developed (Gao et al., 2023; Huang et al., 2021; Li et al., 2022; Qi et al., 2023; Tong et al., 2021; Zhou et al., 2021). However, the models mentioned above are primarily designed for dichotomous data, with only a limited number of methods that can handle polytomous data, those methods involve the normalization of scores from subjective exercises to continuous values in [0, 1] (Gao, Zhao, Li, Zhao, & Zeng, 2022; Liu et al., 2018; Ma, Huang, Tang et al., 2023; Ma, Huang, Zhu et al., 2023; Yang et al., 2022). While polytomous items can be roughly estimated using this method, it collapses several levels (categories) of observed data into a single level of item's difficulty. This represents the ideal situation where the examinee's ability surpasses the item's difficulty, allowing the examinee to answer the item completely correctly. Consequently, this approach often results in a significant loss of information regarding both student ability and item parameters.

To tackle the aforementioned challenges, the most intuitive approach is to either develop a new CDM or extend the existing dichotomous CDMs. When constructing a model for polytomous data, from a design perspective, the main components of deep learning-based CDMs that require modification are the parameters and the multilayer design. As with statistical models, developing a comprehensive polytomous data model poses significant challenges. For example, the polytomous extension of DINA (the Deterministic Input, Noisy 'And' gate) (de La Torre, 2009) and MIRT (Multidimensional item response theory) (Reckase & McKinley, 1991) models require their distinct designs, which will be resource-intensive. Is it possible to process polytomous data directly and apply it to existing CDMs for dichotomous data?

This study aims to accomplish three primary objectives. The first objective is to propose solutions to the problem of existing CDMs extending dichotomous to handle polytomous data by creating a new framework based on data re-encoding. The second objective was to explore methods for preserving parameter interpretability when employing re-encoded data. The third objective is to validate the performance of the proposed framework and assess the impact of increasing the number of score categories on model performance.

To achieve the research objectives, we proposed **P**olytomous **C**ognitive **D**iagnosis **F**ramework (PCDF) which aims to reduce the difficulty of analyzing polytomous data without necessitating the development of entirely new models. The PCDF was created based on features of the response data of ordered polytomous categories. It can be seamlessly integrated with existing dichotomous CDMs, enhancing the estimation accuracy of examinee cognitive state parameters for polytomous data. Additionally, PCDF improves the interpretability of item parameters, achieving the same level of parameter interpretability as polytomous CDMs. The framework consists of three steps. In step 1, items are categorized into appropriate quantities, and dichotomous scores are assigned based on the item's full marks and the examinee's responses employing the **C**umulative **C**ategory **R**esponse **F**unction (CCRF) (Samejima, 1969). In step 2, utilizing the first data division, the data is fed to an existing dichotomous CDM, necessitating adjustment of the item factor parameter to align with the specific model requirements. In step 3, the results of the CDM are merged and recalculated to predict examinees' responses from CCRF. Specifically, PCDF is a general framework and this study applied it to five cognitive diagnosis models (i.e., IRT, MIRT, NCDM, KaNCD, and ICDM) and evaluated against four real-world datasets.

The key contributions are summarized as follows:

1. We proposed PCDF, a general framework for splitting and merging polytomous data using CCRF theory while maintaining model interpretability.
2. PCDF can be seamlessly integrated with existing dichotomous cognitive diagnostic models, reducing the time and resource requirements for new model development.
3. We conducted comprehensive experiments on real-world datasets to validate the performance of PCDF, particularly on interpretability and the effect of increasing the number of score categories.

The subsequent sections of this paper are structured as follows: Section 2 delves into pertinent literature, with a focus on cognitive diagnosis and multi-class classification strategies. Section 3 presents problem formulation and preliminary information about CDMs. Section 4 describes the details of the PCDF. Section 5 reports the results of extensive empirical studies comparing the baselines with PCDF across four real-world datasets. Finally, Section 6 summarizes the work.

2. Related work

This section will introduce three facets of related work: CDMs for dichotomous data, CDMs for polytomous data, and Multi-class classification strategies.

2.1. CDMs for dichotomous data

The earliest cognitive diagnostic models relied on statistical techniques. The standard cognitive diagnostic statistical model known as IRT (Embretson & Reise, 2013) represents trait and logistic function using unidimensional variables. The extended multidimensional model of IRT known as MIRT (Reckase & McKinley, 1991) employs multidimensional vectors to represent the examinees' traits across all dimensions. Another classic model is DINA (de La Torre, 2009), which uses binary variables to assess learners' mastery while also considering the effects of guessing and slipping. These two classic models have been improved upon numerous subsequent models of cognitive diagnostics.

The widespread use of deep learning has sparked an abundance of research in the field of cognitive diagnostics. Wang et al. (2020) proposed a novel NCDM framework, employing neural networks to model the dynamic interaction between learners and responses. The KaNCD (Wang et al., 2022) model was developed by considering the interaction between knowledge concepts, building upon the principles of NCDM. The ICD (Qi et al., 2023) model utilizes interactions among concepts and quantitative relationships between exercises and concepts to improve diagnostic results. The RCD (Gao et al., 2021) model unifies the modeling of interaction and structural relationships through multi-layer student-item knowledge concepts relation maps accurately diagnose students' mastery of specific concepts. Liu, Shen, Qian, and Zhou (2024) proposed ICDM, utilizing student-centered graphs to rapidly diagnose new students' mastery levels.

Furthermore, leveraging the effectiveness of deep learning in handling complex data, subsequent studies have explored various angles to develop cognitive diagnostic models based on deep learning: attribute hierarchy (Li et al., 2022), group level (Huang et al., 2021), context-aware features (Zhou et al., 2021), affect-aware feature (Wang, Zeng, Yang, Xu, & Zhang, 2024), cold start (Gao et al., 2024, 2023), long-tailed problem (Ma et al., 2024; Wang, Zeng, Yang, & Zhang, 2023; Yao et al., 2023), model generalization (Shen, Qian, Zhang, & Zhou, 2024), integrate with knowledge tracing (Su et al., 2024), longitudinal assessment (Huang et al., 2024), and multi-modal data (Song et al., 2023). However, these models were intended to work exclusively with dichotomous scored items, like math objective questions.

2.2. CDMs for polytomous data

Consider a real-world scenario, a quiz may have a wide range of question types, including objective, subjective, essay, and Likert-type items. It has also resulted in a variety of scoring methods, with dichotomous and polytomous items coexisting on the same quiz, rendering dichotomous CDMs ineffective. To address this challenge, researchers have proposed several cognitive diagnostic models that employ polytomous items.

With the aid of a cumulative category response function (Samejima, 1995) based on the dichotomous response model (like IRT), Samejima (1969) created the graded response model (GRM), which is widely used. Many scholars have developed polytomous response models based on this theory, including MGRM (multidimensional graded response model) (de Ayala, 1994) based on MIRT, P-LCDM (Henson, Templin, & Willse, 2009) based on LCDM, and GPDM (Chen & de la Torre, 2018) based on G-DINA.

In addition, deep learning-based CDMs make an effort to handle data with polytomous items. FuzzyCDF (Liu et al., 2018) fuzzifies the examinees' skill proficiency to handle the partially correct responses to subjective problems. Specifically, it redefines the polytomous score as fuzzy fractions of value in [0,1]. The majority of the follow-up CDMs that dealt with subjective problems were approached in this way, such as SPP-NCD (Ma, Huang, Tang et al., 2023), FC-CDF (Ma, Huang, Zhu et al., 2023), deepCDF (Gao et al., 2022), and QRCDM (Yang et al., 2022). However, this processing only yields approximate information about the examinees' ability, lacks accurate prediction of the student's score, and omits information about the question's parameters.

2.3. Multi-class classification strategies

Typically, constructing a classifier to distinguish between two classes is easier than considering more than two classes in a single problem, as the decision boundaries tend to be simpler in the former case (Galar, Fernández, Barrenechea, Bustince, & Herrera, 2011). Binarization techniques address multi-class problems by partitioning the original task into simpler binary classification sub-problems handled by binary classifiers. Common decomposition strategies include one-vs-all and one-vs-one approaches.

The one-vs-all method (Anand, Mehrotra, Mohan, & Ranka, 1995) trains a classifier for each class, wherein each class is distinguished from all other classes, such that a positive output from the base classifier indicates the corresponding class. The one-vs-one method (Knerr, Personnaz, & Dreyfus, 1990) trains a binary classifier for every pair of classes, yielding a distinct classifier for each possible pair of classes present in the dataset.

However, these multi-class classification strategies are not directly applied to CDMs. In particular, CDMs are often designed with interpretable parameters, and it may result in the interpretability of original parameters lost in CDMs even if these strategies can be used. In this study, the theory of the one-vs-all method as a comparison strategy with PCDF. To be more precise, every ordinal multi-classification item is re-encoded into multiple dichotomous classifications, and assigns a score of 1, with 0 applied in the other cases to each re-encoded item.

3. Preliminary

3.1. Research background

Here is a formal definition of cognitive diagnosis. Suppose there are N examinees, M Items, and K dimensions in a test, which can be represented as $S = \{s_1, s_2, \dots, s_N\}$, $E = \{e_1, e_2, \dots, e_M\}$, and $D = \{d_1, d_2, \dots, d_K\}$, respectively. Each examinee will finish all items, and the response logs R are denoted as a set of triplet (s_i, e_j, r_{ij}) where $s_i \in S$, $e_j \in E$, and $r_{ij} \in \{0, 1, \dots, f_j\}$ is the specific score of examinee s_i in item e_j, where f_j is the full mark of item e_j. In addition, the test has a Q-matrix $Q = \{Q_{jk}\}_{M\times K}$, where $Q_{jk} = 1$ if item e_j relates to dimension d_k and $Q_{jk} = 0$ otherwise.

Problem Definition: Given the examinees' response logs R and the Q-matrix Q, our goal is to develop a data transformation method that enhances cognitive diagnostic model performance when handling polytomous data.

3.2. Dichotomous cognitive diagnosis models

Given that our method must be used in conjunction with an existing dichotomous cognitive diagnostic model, we will begin by offering a concise introduction to dichotomous CDMs. The CDMs are the tools utilized to evaluate examinees' mastery of particular dimensions (knowledge concepts) by analyzing their responses to various test items,i.e., $\Theta, \Phi \leftarrow R$, where Θ is the latent abilities of examinees, Φ is the item-related parameters, and R is the responses data. The examinee performance prediction task is commonly employed to acquire Θ and Φ, with the optimization goal of minimizing the discrepancy between the true response r_{ij} and the predicted probability $P(y_{ij})$. Cross entropy is a frequently used loss function:

$$loss = -\sum_{i,j}(r_{ij}\log y_{ij} + (1-r_{ij})\log(1-y_{ij})) \tag{1}$$

Five commonly used CDMs are described as follows.

- **IRT** (Embretson & Reise, 2013) uses a logistic-like function to simulate the features of unidimensional items and examinees. The correct probability of examinee s_i answering item e_j is usually modeled with the classical IRT metric:

$$P(Y_{ij} = 1|\theta_i) = \frac{1}{1+\exp\{-1.7a_j(\theta_i - b_j)\}}, \tag{2}$$

  where θ_i denotes the ability level of the examinee s_i, and a_j and b_j indicate the discrimination and difficulty of item e_j.
- **MIRT** (Reckase & McKinley, 1991) is a multidimensional version of IRT that simulates examinees' multiple knowledge concepts abilities through items:

$$P(Y_{ij} = 1|\Theta_i) = \frac{1}{1+\exp\{-1.7(\mathbf{a}_j^T\Theta_i - b_j)\}}. \tag{3}$$

  where $\mathbf{a}_j$ and Θ_i are multidimensional vectors representing the ability level of the examinee s_i and discrimination of item e_j, respectively.
- **NCDM** (Wang et al., 2020) is one of the most widely used deep learning-based CDMs, it employs a multilayer perceptron (MLP) to capture intricate and high-order examinee-item interaction functions:

$$P(Y_{ij} = 1|\Theta_i) = MLP(\mathbf{Q}_e \circ (\mathbf{h}_s - \mathbf{h}^{diff}) \times h^{disc}), \tag{4}$$

  where $\mathbf{h}^{diff}$ and h^{disc} denote knowledge difficulty and item discrimination, $\mathbf{h}_s$ represents the examinee's knowledge proficiency, and MLP contains two full connection layers and an output layer.
- **KaNCD** (Wang et al., 2022) builds upon the advancements of NCDM and incorporates implicit relationships between knowledge attributes.

$$P(Y_{ij} = 1|\Theta_i) = MLP(\mathbf{Q}_e \circ (\mathbf{h}_s \cdot \mathbf{l} - \mathbf{h}^{diff} \cdot \mathbf{l}) \times h^{disc}), \tag{5}$$

  where $\mathbf{l}$ denotes the knowledge concept vector associated with the item e.
- **ICDM** (Liu et al., 2024) is a CDM that leverages aggregated results from students' neighbors in student-centered graphs to quickly assess incoming examinees' mastery levels.

$$P(Y_{ij} = 1|\Theta_i) = \sigma(MLP((\mathbf{MAS}_{s_i} - \mathbf{Diff}_{e_j}) \odot \mathbf{Q}_{e_j})), \tag{6}$$

  where $\mathbf{MAS}_{s_i}$ is the mastery level of examinee s_i, $\mathbf{Diff}_{e_j}$ is the difficulty of item e_j, σ represents the Sigmoid function.

3.3. The cumulative category response function

Given that the MIRT was considered for designing the NCDM, it is necessary to refer to the statistics-based polytomous models of IRT and CD for the PCDF proposed in this study. The general graded-response model (GGRM) (Samejima, 1969) is a cluster of graded-response models (GRM) designed based on the cumulative category response function (CCRF) and is widely used in education

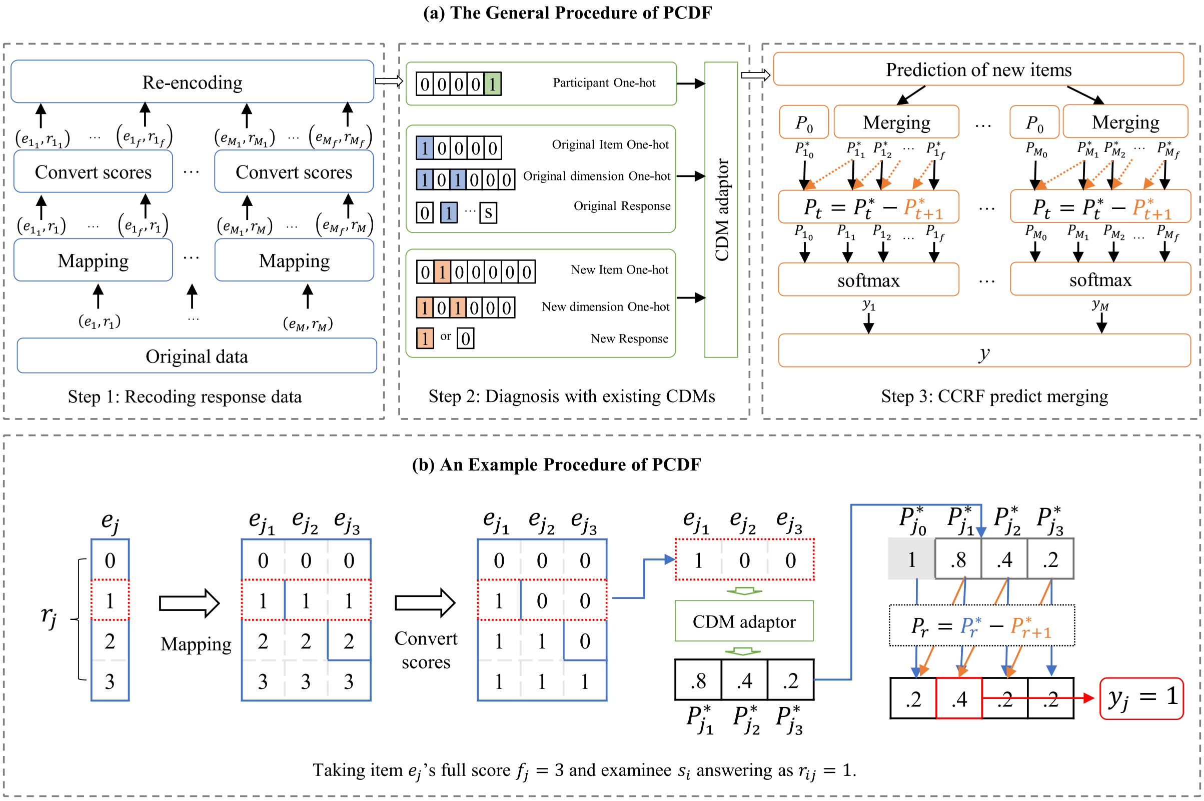


Fig. 1. The framework of the proposed PCDF: (a) The general procedure; (b) An example procedure.

to process polytomous data. The following section will introduce GGRM and CCRF, and explain why the CCRF can be applied to data re-encoding of polytomous data.

The general graded-response model is a set of mathematical models used to analyze ordered polytomous categories. Samejima (1995) proposed that a cognitive process consists of a finite number of steps. For an examinee scoring t be assigned that the examinee can complete up to step t, but fail to complete the step $t+1$. Let $M_t(\theta)$ be the processing function which the examinee completes the step t successfully.

The category response function $P_t(\theta)$ of the graded item score t is expressed as

$$P_t(\theta) = \left[\prod_{z \le t} M_z(\theta)\right]\left[1 - M_{(t+1)}(\theta)\right], \tag{7}$$

where θ is the latent trait of the examinee, let $P_t^*(\theta) = \prod_{z \le t} M_z(\theta)$ be the conditional probability that the examinee successfully completes the cognitive process up to step t or beyond. CCRF can be expressed as

$$P_t(\theta) = P_t^*(\theta) - P_{t+1}^*(\theta). \tag{8}$$

Let f be the full mark for the item, it is reasonable to constrain $P_{t=0}^*(\theta) \equiv 1$ and $P_{t=f+1}^*(\theta) \equiv 0$. When transforming the graded-item score t into a binary score, we assign 0 to all score categories less than t and 1 to those equal to or greater than t.

Selecting an interpretable model requires certain desirable properties, one of which is the additively of the score category response function. Furthermore, even after more precise recategorizations or combinations of two or more categories, the score category functions stay within the same mathematical model.

In this study, the additivity of the score category response functions is used in the processing of the response data.

4. Polytomous neural cognitive diagnosis framework

Polytomous data is processed by the polytomous cognitive diagnosis framework (PCDF) using the cumulative category response function. Fig. 1 illustrates the PCDF.

4.1. Step 1: Re-encoding response data

Each item in the original data is implicitly mapped to new items based on the number of scoring categories for the original item. For the original item e_j, the corresponding sequence of new items are

$$T_{e_j} = \{e_{j_1}, e_{j_2}, \dots, e_{j_{f_j}}\}, \tag{9}$$

where f_j is the full mark for item j, the number of T_{e_j} is equal to f_j. In other words, graded-item score $f_j + 1$ item can be divided into f_j binary score items. As shown in the procedure section of Fig. 1, the score categories of item e_j are $r_j \in \{0, 1, 2, 3\}$, encoding item e_j as item e_{j_1}, item e_{j_2}, and item e_{j_3}. If $r_j \geq 1$, $r_{j_1} = 1$ and $r_{j_1} = 0$ otherwise; if $r_j \geq 2$, $r_{j_2} = 1$ and $r_{j_2} = 0$ otherwise; if $r_j \geq 3$, $r_{j_3} = 1$ and $r_{j_3} = 0$ otherwise.

Thus, the new items set after re-encoding is

$$T = \{T_{e_1}, T_{e_j}, \dots, T_{e_M}\}, \tag{10}$$

the total number of new items is

$$L = \sum_{j=1}^{M} f_j. \tag{11}$$

In order to make it more easily expressed, we rephrase the new item set as

$$T = \{t_1, t_2, \dots, t_l, \dots, t_L\}, \tag{12}$$

where t_l is the lth item in the new items set. After converting scores, the new items answer $r_{il} \in \{0, 1\}$. In addition, the tests have a new Q-matrix $Q = \{Q_{lk}\}_{L \times K}$, where $Q_{lk} = 1$ if item t_l relates to dimension d_k and $Q_{lk} = 0$ otherwise. The dimension of the new item is mapped directly from the corresponding original item.

One-hot representation vector. As for each examinee, their one-hot representation vectors $\mathbf{x}^s \in \{0, 1\}^{1 \times N}$ are not re-encoded. As for each original item, the one-hot representation vector of the dimensions contained in the item is given by

$$\mathbf{Q}_e = \mathbf{x}^e \times \mathbf{Q}, \tag{13}$$

where $\mathbf{Q}_e \in \{0, 1\}^{1 \times K}$, $\mathbf{x}^e \in \{0, 1\}^{1 \times M}$ is the one-hot representation of the original item. After the item is re-encoded, Eq. (13) will change to

$$\mathbf{Q}_t = \mathbf{x}^t \times \mathbf{Q}, \tag{14}$$

where $\mathbf{Q}_t \in \{0, 1\}^{1 \times K}$, $\mathbf{x}^t \in \{0, 1\}^{1 \times L}$ is the one-hot representation of the new item.

To summarize, the outcomes of re-encoding the original data are the examinee, original item, original dimension, new item, and new dimension one-hot representation vector are $\mathbf{x}^s$, $\mathbf{x}^e$, $\mathbf{Q}_e$, $\mathbf{x}^t$, and $\mathbf{Q}_t$, respectively, as well as the original and new response. In Step 2, the CDM adaptor will use them as optional input and output parameters.

4.2. Step 2: Diagnosis with existing CDMs

The prediction formula for examinees' responses to the new items are

$$P_T^* = CDM(\theta, \Phi_T), \Phi_T = \{a, b, \dots\}, \tag{15}$$

where CDM represents existing cognitive diagnosis models such as IRT, MIRT, NCDM, KaNCD, and ICDM; Φ_T represents the item-related parameters of the traditional cognitive diagnostic model used (e.g., item discrimination a, item difficulty b). Note that the meaning of the parameters has changed due to the re-encoding of the original data in Step 1. For item discrimination a, the number of it in the polytomous data was no different from the dichotomous data. Therefore, when coding the item discrimination a, the one-hot representation vector $\mathbf{x}^e$ was used. However, the polytomous data have more than one item difficulty. Because of the re-encoding rules, the new item difficulty represents the difficulty of the examinees scoring different points on the original items, the new item difficulty is coded using the one-hot representation vector of new items. That is, for item difficulty b, b_l was used instead of b_j, and the new item one-hot representation vector $\mathbf{x}^t$ was used.

The loss function is cross entropy between the output:

$$loss_{CDM} = Entropy(P_T^*, R_T), \tag{16}$$

where R_T denotes the response labels for the examinee in new items.

4.3. Step 3: CCRF predicts merging

What is obtained through Step 2 is an estimate of the probability of responding to the new items, which is not the final estimate for examinees on the original items. In Step3, the probability of the new items is merged by mapping the new items to the original items in Step 1:

$$P_T^* = \{P^*_{T_{e_1}}, P^*_{T_{e_2}}, \dots, P^*_{T_{e_M}}\}, \tag{17}$$

$$P^*_{T_{e_j}} = \{P^*_{e_{j_1}}, P^*_{e_{j_2}}, \dots, P^*_{e_{j_{f_j}}}\}, \tag{18}$$

where $P_0 \equiv 1$ represents the probability that the examinee scores 0 and above is always equal to 1.

The probability that the examinee choosing each score according to Eq. (8):

$$P = \{P_{r=0}, P_{r=1}, \dots, P_{r=f}\}. \tag{19}$$

Considering that all the probabilities should add up to 1, using the softmax function for P,

$$p_t = softmax(P_t) = \frac{\exp(P_t)}{\sum_{t=0}^{f}\exp(P_t)}, \tag{20}$$

the corresponding score in p with the largest probability value is the final predicted score

$$y = max(p). \tag{21}$$

An example is given in Fig. 1(b). After applying the CDM adaptor, we obtain the probabilities of answering items e_{j_1}, e_{j_2}, and e_{j_3} correctly, corresponding to $P^*_{j_1}$, $P^*_{j_2}$, and $P^*_{j_2}$. Since $P^*_{j_0} \equiv 1$, applying Eq. (8) and (20) yields the probabilities of scoring each mark on e_j as [0.2, 0.4, 0.2, 0.2]. Here, the probability is maximized at 0.4, so the corresponding score should be $y_j = 1$.

5. Evaluation

As the primary contribution of this study is to expand on current cognitive diagnosis models to accommodate adaptively polytomous data, we evaluate the effectiveness of CDMs combined with our PCDF and the combined linear-split and one-vs-all data re-encoded methods (hereinafter referred to as PCDF-CDMs, LS-CDMs, and OVA-CDMs, respectively) to answer the following questions:

- **RQ1:** Can PCDF-CDMs able to perform better than LS-CDMs and OVA-CDMs?
- **RQ2:** How about the interpretability of PCDF diagnostic examinee states?
- **RQ3:** How about the PCDF diagnostic item difficulty's interpretability?
- **RQ4:** The visualization interactions between examinees, items, and dimensions.

5.1. Experimental setting

5.1.1. Dataset description

There are two sources of open real-world datasets,[1] i.e., the Cattell's 16 Personality Factors Test (16PF) (Cattell & Mead, 2008) and the Fisher Temperament Inventory (FTI) (Brown, Acevedo, & Fisher, 2013).

16PF is a total of 163 items, consisting of 16 factors, presented on a 5-point Likert scale. Examinees are asked to rate their approval of each item. The 16PF dataset was preprocessed using the following criteria: accuracy of 95–100, age below 70, response time between 300 and 1200s, data with a missing rate of less than 10%, and number of continuous responses less than 10. Finally, 14,921 examinees' responses were kept.

The original 16PF is a 5-point Likert scale. To explore the impact of the number of points on the model performance, the formal dataset was divided into 3 parts: (**1**) The dataset 1 is combined into a 3-point data, named 16PF-g3, (**2**) The dataset 2 is unchanged, named 16PF-g5, and (**3**) The dataset 3 is a mixed score that combined from 2-point to 5-point data, named 16PF-gmix. The combined scoring rules are shown in Table 1.

FTI is a total of 56 items, consisting of 4 factors, presented on a 4-point Likert scale. Examinees are asked to rate their approval of each item. The FTI dataset was preprocessed using the following criteria: age below 70, a response time between 300 and 1200 s, data with a missing rate of less than 10%, and several continuous responses of less than 10. Finally, 4412 examinees were left. Table 2 summarizes the basic statistics of the datasets.

[1] They are from a website named Open-Source Psychometrics Project: https://openpsychometrics.org/_rawdata/.

Table 1
Rules of combined grade.

	1	2	3	4	5
2-point	1	1	1	2	2
3-point	1	1	2	3	3
4-point	1	1	2	3	4

Table 2
Dataset summary.

	16PF-g3	FTI	16PF-g5	16PF-gmix
Factors	16	4	16	16
Examinees	4974	4412	4974	4973
Items	163	56	163	163
Items for split-grade	326	168	652	406
Number of 2-point	–	–	–	41
Number of 3-point	163	–	–	41
Number of 4-point	–	56	–	41
Number of 5-point	–	–	163	40
Factors per item	1	1	1	1
Response logs	808 759	247 072	808 573	808 437

5.1.2. Baselines

To validate the efficacy of PCDF, we utilized five standard diagnostic methods as CDM adaptors in Step 2, namely IRT, MIRT, NCDM, KaNCD, and ICDM, which we refer to as P-IRT, P-MIRT, P-NCDM, P-KaNCD, and P-ICDM, respectively. These methods serve as the foundation for a multitude of CDM models and have undergone extensive validation in numerous studies.

Three baselines were chosen for comparison. Among them, the random indicates the lower bounds of performance. The specifics are outlined below:

- **Linear-Split (LS)**. The model was trained by normalizing the scores as continuous values in [0, 1] (Gao et al., 2022; Liu et al., 2018; Ma, Huang, Tang et al., 2023; Ma, Huang, Zhu et al., 2023; Yang et al., 2022). After training the CDM adaptor, the result obtained is the probability of predicting the examinee's full score. This probability is then linearly transformed to produce the polytomous score. The conversion depends on the number of score categories. For example, the full mark of item e_j is 3; the corresponding score category intervals are $p_{ij} \in [0, 0.25) \rightarrow \hat{y}_{ij} = 0$, $p_{ij} \in [0.25, 0.5) \rightarrow \hat{y}_{ij} = 1$, $p_{ij} \in [0.5, 0.75) \rightarrow \hat{y}_{ij} = 2$, and $p_{ij} \in [0.75, 1] \rightarrow \hat{y}_{ij} = 3$. LS, derived directly from existing CDMs, is designated as L-IRT, L-MIRT, L-NCDM, L-KaNCD, and L-ICDM, corresponding to IRT, MIRT, NCDM, KaNCD, and ICDM, respectively.
- **One-vs-all (OVA)** (Anand et al., 1995). Each ordinal multi-classification item is re-encoded into $f + 1$ classifications dichotomous items. Each re-encoding item sets one of the scores to 1 and 0 otherwise. The prediction for the re-encoded items is combined to get the prediction for the original items. To illustrate the OVA method, consider an example in which item e_j has a full score of 3 and examinee s_i scores 1 on e_j. Item e_j can be re-encoded as $\{e_{j_0}, e_{j_1}, e_{j_2}, e_{j_3}\}$, with the corresponding convert scores of $\{0, 1, 0, 0\}$. After training the model with the CDM adaptor, $\{p_{j_0}, p_{j_1}, p_{j_2}, p_{j_3}\}$ represent the examinee's probabilities of receiving each score on e_j. The final score will be the subscript of j representing the highest probability value. The combination of OVA with IRT, MIRT, NCDM, KaNCD, and ICDM are denoted as O-IRT, O-MIRT, O-NCDM, O-KaNCD, and O-ICDM, respectively.
- **Random**. The random method predicts examinees' scores randomly from a uniform distribution $Uniform(0, 1)$, with the scores handled following LS.

5.1.3. Experimental details

The response data from each examinee was randomly split into training and test sets in an 8:2 ratio. To ensure fairness, 90% of the training set was utilized as training data, while the remaining 10% was allocated as validation data for adjusting hyperparameters. All experiments were conducted 10 times, and the average of the results from the 10 repetitions was considered as the final experimental outcome.

We use Adam Optimizer to tune all the methods to their best performance. The k-hop k in ICDM was optimized from the set $\{1, 2, 3, 4\}$. The learning rate and batch size are chosen from the sets $\{5e{-}3, 2e{-}3, 1e{-}3, 5e{-}4\}$ and $\{32, 64, 128\}$, respectively. The other parameters for IRT, MIRT, NCDM, and KaNCD were determined using guidance from EduCDM,[2] while the other parameters for ICDM were determined with guidance from ICDM.[3] All experiments are run on a Linux server with a 2.30 GHz Intel(R) Xeon(R) Gold 5218 CPU and RTX A6000 GPU; the running memory is 48 GB.

[2] https://github.com/bigdata-ustc/EduCDM.
[3] https://github.com/ECNU-ILOG/ICDM.

Table 3
Experimental results of examinee performance prediction on the 16PF-g3 and FTI datasets. (Note: As Random generates predictions randomly and LS does not involve data re-encoding, neither of them possesses evaluation metrics for step 2. The following table follows the same convention.)

CDM adaptor	Method	16PF-g3					FTI				
		ACC_d	RMSE	AUC	ACC_p	WACC	ACC_d	RMSE	AUC	ACC_p	WACC
	Random	–	–	–	0.333	0.531	–	–	–	0.250	0.523
RT	LS	–	–	–	0.479	**0.705**	–	–	–	0.460	0.712
	OVA	**0.737**	0.424	0.754	0.589	0.679	0.752	0.406	0.730	0.437	0.699
	PCDF	0.712	**0.433**	**0.766**	**0.604**	0.697	**0.798**	**0.376**	**0.870**	**0.465**	**0.712**
MIRT	LS	–	–	–	0.612	**0.763**	–	–	–	0.493	0.729
	OVA	0.737	0.424	0.755	0.588	0.679	0.752	0.406	0.730	0.436	0.699
	PCDF	**0.764**	**0.405**	**0.827**	**0.653**	0.753	**0.817**	**0.361**	**0.890**	**0.505**	**0.738**
NCDM	LS	–	–	–	0.612	**0.764**	–	–	–	0.495	0.732
	OVA	0.756	0.416	0.768	0.635	0.728	0.750	0.406	0.731	0.439	0.698
	PCDF	**0.763**	**0.405**	**0.827**	**0.650**	0.754	**0.815**	**0.363**	**0.889**	**0.505**	**0.737**
KaNCD	LS	–	–	–	0.623	**0.766**	–	–	–	0.499	**0.734**
	OVA	0.747	0.421	0.755	0.598	0.690	0.751	0.407	0.729	0.437	0.699
	PCDF	**0.771**	**0.396**	**0.837**	**0.661**	0.758	**0.815**	**0.361**	**0.890**	**0.499**	0.734
ICDM	LS	–	–	–	0.628	**0.769**	–	–	–	0.498	0.734
	OVA	0.760	0.414	0.743	0.600	0.691	0.751	0.409	0.722	0.425	0.686
	PCDF	**0.768**	**0.398**	**0.836**	**0.659**	0.755	**0.816**	**0.360**	**0.892**	**0.502**	**0.736**

5.1.4. Evaluation metrics

Evaluating the performance of a CDM proves challenging due to the unknown latent traits of the examinees. Following previous works (Gao et al., 2022; Wang et al., 2020), a common approach is to evaluate performance based on prediction scores in diagnostic models. Different evaluation metrics were used for Step 2 and Step 3.

In Step 2, accuracy (ACC), root mean square error (RMSE), and area under the curve (AUC) were used in the CDM adaptor to evaluate prediction performance on re-encoding data. In Step 3, ACC and the weighted accuracy (WACC) were used as evaluation metrics for the performance of final score prediction. To better distinguish the difference between ACC in step 2 and step 3, we name them ACC_d and ACC_p respectively. WACC references the weighted classification method (Chen & de la Torre, 2013) which is more appropriate for the assessment of classification accuracy for polytomous attributes (Chen & de la Torre, 2013; Zhan, Wang, & Li, 2020). WACC can provide a better assessment of misclassification degree than ACC. For instance, under the WACC method, a true score of 2 estimated as 1 is considered superior to estimating it as 0, whereas both estimates are equally deemed incorrect under the ACC method.

$$ACC_p = \frac{1}{\sum_{i=1}^{N} |E_i^Y|} \sum_{i=1}^{N} \sum_{e_j \in E_i^Y} I(y_{ij}, \hat{y}_{ij}), \tag{22}$$

$$W_{ij} = \begin{cases} 0.5^{|y_{ij} - \hat{y}_{ij}|}, & \text{if } |y_{ij} - \hat{y}_{ij}| < f_j \\ 1, & \text{otherwise} \end{cases} \tag{23}$$

$$WACC = \frac{1}{\sum_{i=1}^{N} |E_i^Y|} \sum_{i=1}^{N} \sum_{e_j \in E_i^Y} W_{ij} \tag{24}$$

where y_{ij} is the examinee s_i's answer label to item e_j, $\hat{y}_{ij}$ is its estimate label.

5.2. Examinee performance prediction comparison (RQ1)

Tables 3 and 4 shows the examinees' score predictions for baseline models and the PCDF; the best results are shown in bold. We have some findings. First, across various diagnostic implementations (i.e., IRT, MIRT, NCDM, KaNCD, and ICDM as CDM adaptor), our proposed PCDF framework demonstrates superior performance compared to most baseline models (including Random), underscoring the effectiveness of PCDF in predicting examinee performance in polytomous data. Second, among the various baseline models, demonstrate superior performance compared to the IRT baseline. This suggests that employing multidimensional trait vectors of items and examinees enables more accurate capture of information about examinees' abilities. Third, on the 16PF-g3, FTI, and 16PF-g5 datasets, the accuracy of the predictions for examinees decreased as the number of score categories increased. Since the 16PF-gmix dataset includes items with different numbers of score categories, it does not reflect this pattern. However, the results of the experiment on those datasets show that PCDF has the ability to process items with different numbers of score categories simultaneously.

To better understand the variation in the predictive performance of models across the various numbers of item categories, we conducted a categorical analysis of prediction evaluation on items with varying numbers of graded scores in 16PF-gmix dataset. Figs. 2 and 3 show the results of ACC_p and WACC for the different number of graded scores in 15 conditions, respectively. Results are as follows: (**1**) A consistent decrease in ACC_p across all five CDMs under the LS strategy as the number of score levels increases

Table 4
Experimental results of examinee performance prediction on the 16PF-g5 and 16PF-gmix datasets.

CDM adaptor	Method	16PF-g5					16PF-gmix				
		ACC_d	RMSE	AUC	ACC_p	WACC	ACC_d	RMSE	AUC	ACC_p	WACC
	Random	–	–	–	0.200	0.451	–	–	–	0.322	0.497
IRT	LS	–	–	–	0.347	0.625	–	–	–	0.467	0.662
	OVA	**0.800**	**0.382**	0.707	0.375	0.617	0.753	0.413	0.746	0.511	0.650
	PCDF	0.787	0.384	**0.862**	**0.395**	**0.631**	**0.749**	**0.411**	**0.826**	**0.531**	**0.663**
MIRT	LS	–	–	–	0.436	0.678	–	–	–	0.560	**0.715**
	OVA	0.800	0.382	0.707	0.374	0.617	0.753	0.413	0.746	0.511	0.650
	PCDF	**0.821**	**0.358**	**0.894**	**0.466**	**0.686**	**0.788**	**0.386**	**0.866**	**0.586**	0.710
NCDM	LS	–	–	–	0.432	0.679	–	–	–	0.560	0.716
	OVA	0.800	0.382	0.712	0.376	0.618	0.761	0.408	0.762	0.552	0.685
	PCDF	**0.821**	**0.360**	**0.894**	**0.464**	**0.689**	**0.787**	**0.385**	**0.867**	**0.583**	**0.719**
KaNCD	LS	–	–	–	0.434	0.679	–	–	–	0.560	**0.720**
	OVA	0.800	0.383	0.707	0.374	0.617	0.754	0.415	0.746	0.512	0.649
	PCDF	**0.823**	**0.355**	**0.898**	**0.470**	**0.690**	**0.794**	**0.378**	**0.874**	**0.594**	0.717
ICDM	LS	–	–	–	0.442	0.684	–	–	–	0.559	**0.719**
	OVA	0.800	0.384	0.706	0.374	0.616	0.752	0.414	0.745	0.511	0.649
	PCDF	**0.822**	**0.355**	**0.898**	**0.460**	**0.685**	**0.793**	**0.379**	**0.873**	**0.591**	0.714

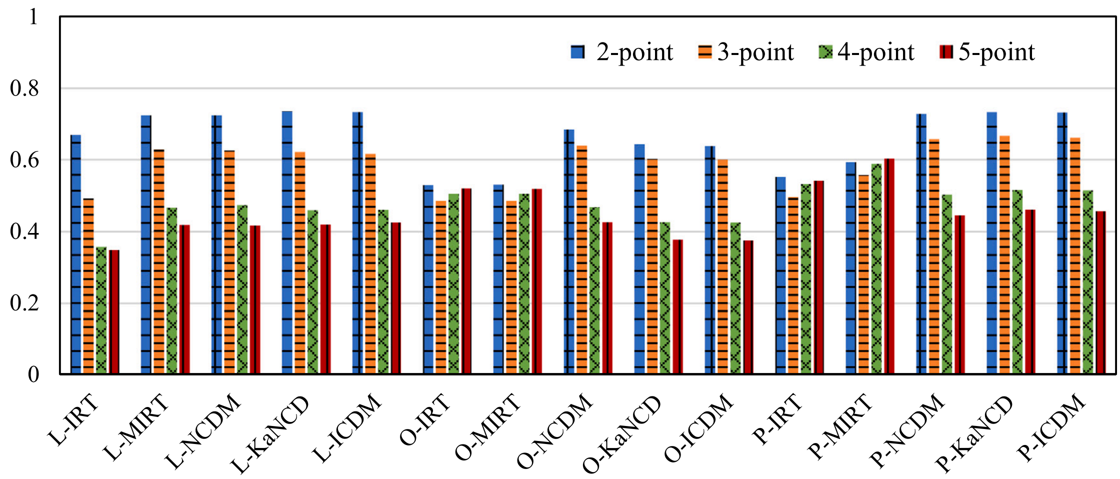


Fig. 2. ACC_p results for different score points in the 16PF-gmix dataset.

and for OVA and PCDF strategies, a similar trend is observed in NCDM, KaNCD, and ICDM, with PCDF demonstrating higher ACC_p and greater interpretability. (**2**) Similar patterns are observed in WACC metrics, but the decline is smaller than the ACC_p because the WACC weights incorrect responses that are in the vicinity of the correct answer. (**3**) In both IRT and MIRT in OVA and PCDF strategies, there is minimal variance in ACC_p across different numbers of graded scores, with an upward trend in WACC. In summary, the increase in the number of graded scores affects the model performance and the model performs better under the PCDF strategy.

5.3. Explainability of diagnostic examinee states (RQ2)

To evaluate the cognitive diagnosis models, more than just the examinee performance prediction task is required, because interpretability is an important aspect of cognitive diagnostic results.

In particular, the diagnosis examinee states were evaluated using the Criterion Validity (CV) (Campbell & Fiske, 1959) as the evaluation metric. This metric is a reference standard to measure the effectiveness of the test, which is independent of the test and can reflect the purpose of the test. It is often used in psychological tests. The degree of correlation between test score and criterion is expressed by a correlation coefficient, which is called the validity coefficient. The larger the validity coefficient, the higher the validity of the test.

In this study, the Criterion Validity was used to verify the validity of the examinees' dimensional state results. The 2PL-GRM model was chosen as the validity model, where EM (expectation maximization algorithm) was used for item parameters estimation and EAP (expected a posteriori) was used for examinee states estimation. Pearson's correlation coefficient (Pearson, 1900) was used for the solution.

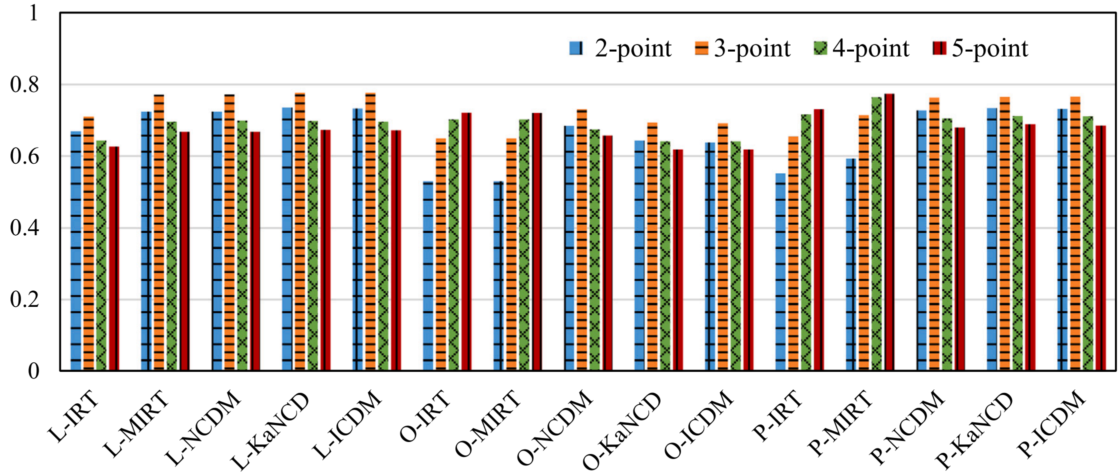


Fig. 3. WACC results for different score points in the 16PF-gmix dataset.

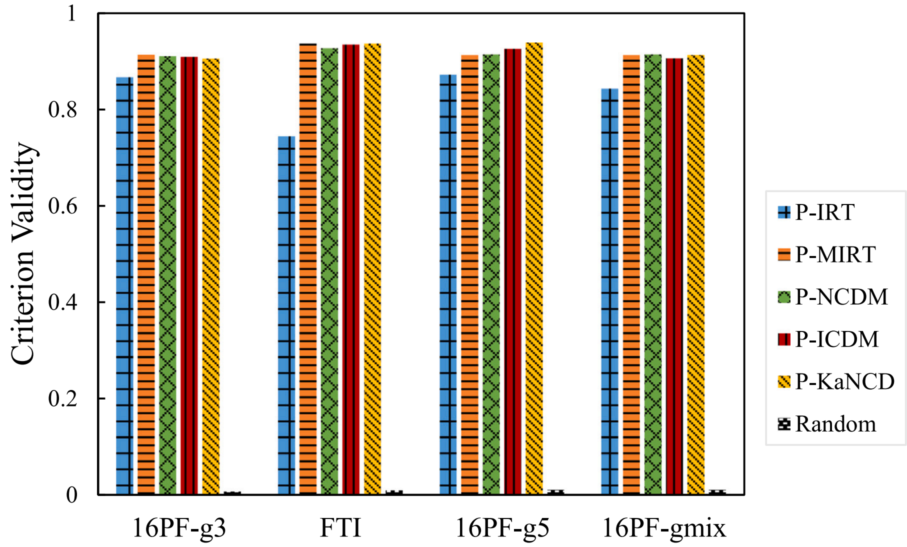


Fig. 4. CV results of models on four datasets.

CV is formulated as:

$$CV = \frac{1}{K}\sum_{k=1}^{K} corr(h_k, \theta_k), \tag{25}$$

where K is the total number of dimensions of datasets, h_k is the estimate of the latent traits of all examinees in the kth dimension, and θ_k is the corresponding estimated using 2PL-GRM as criterion value.

Fig. 4 presents the CV comparison results among IRT, MIRT, NCDM, KaNCD, ICDM, and Random in PCDF across four datasets. First, all the models demonstrate significantly greater effectiveness compared to the Random model. Second, the CVs of P-IRT were lower than those of the other models (above 0.74), while the CVs for the remaining models were very close (exceeding 0.9). Finally, the models with the highest interpretability across the four datasets were P-MIRT, P-KaNCD, P-KaNCD, and P-NCDM. This result demonstrates the high interpretability of the ability parameters obtained from the PCDF.

5.4. Explainability of diagnostic difficulty results (RQ3)

To illustrate the impact of LS, OVA, and PCDF methods on the interpretability of item parameters, we analyze their item difficulty parameter values using KaNCD as a CDM adaptor in Fig. 5. The findings are as follows: (**1**) The LS method assigns each item a single difficulty parameter indicating the challenge of achieving full marks. While the LS method estimates the examinee’s ability value,

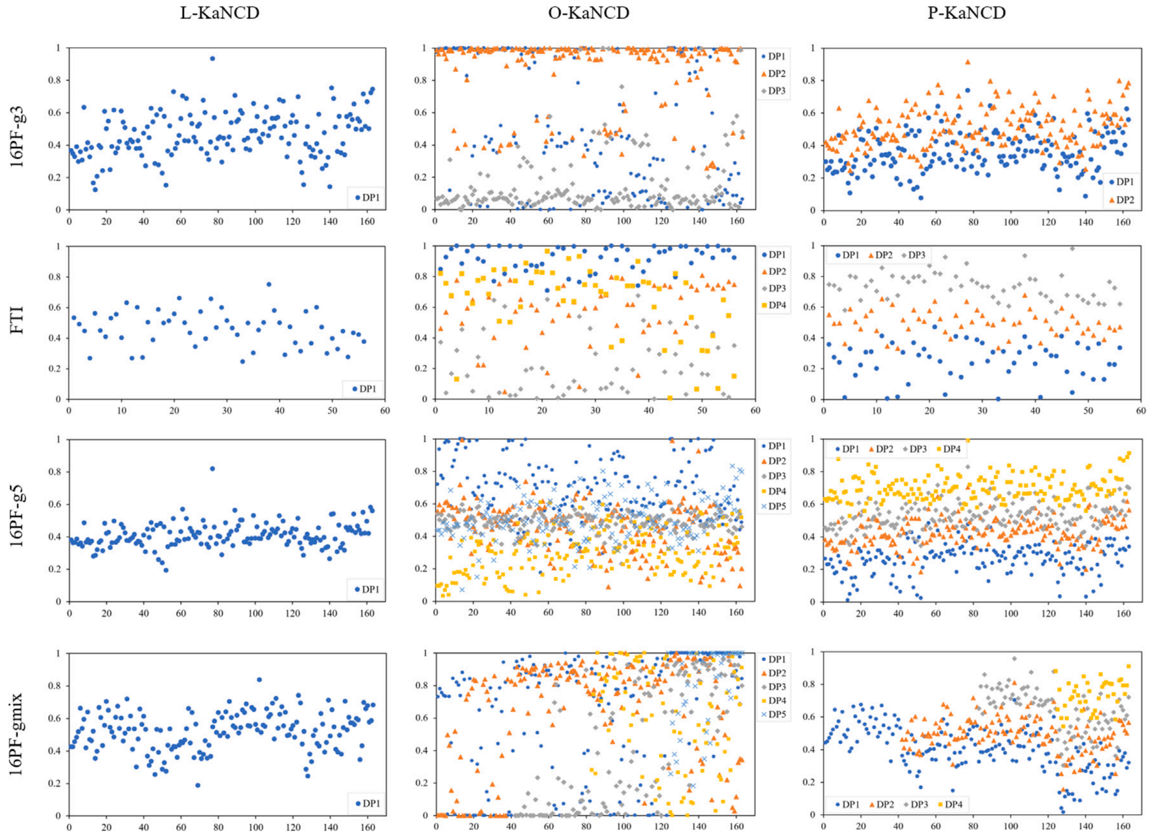


Fig. 5. Difficulties in parameter distribution of P-KaNCD on four datasets. To better present the distribution of difficulty parameters, the items in the dataset 16PF-gmix were reordered according to the number of score categories.

it does not accurately predict their score. (**2**) The OVA method shows no distinct pattern in item difficulty parameters as it divides polytomous items solely based on graded levels, leading to ambiguous difficulty parameters. (**3**) In contrast, the PCDF method assigns multiple difficulty parameters to polytomous items, which increase monotonically. This implies that higher examinee ability correlates with higher item scores. Moreover, items with the same score exhibit similar difficulty levels, which increase with the score. This underscores the interpretability of PCDF-derived difficulty parameters and their adherence to the principle of monotonic increase.

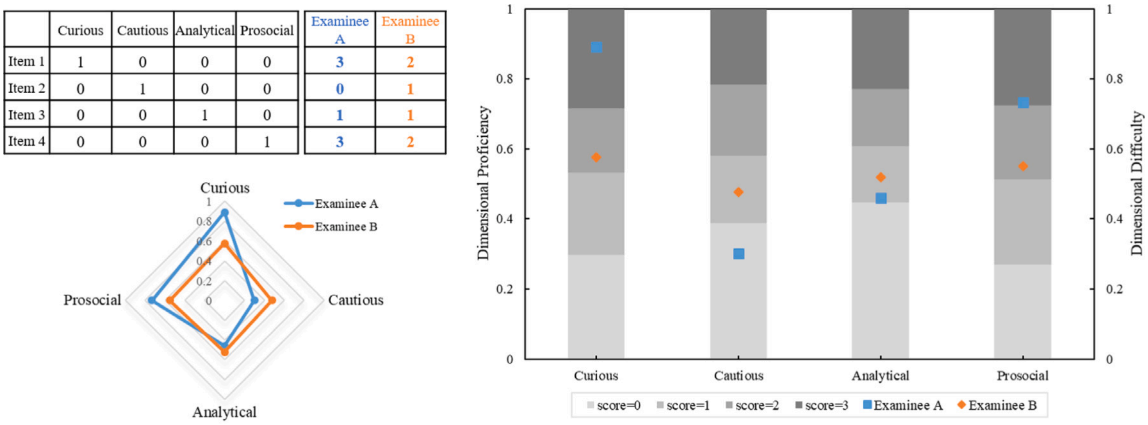
5.5. Case study (RQ4)

Fig. 6 presents an example of two examinees' diagnostic results by P-NCD on the public dataset FTI. The table in Fig. 6(a) displays the Q-matrix for four items answered by examinees A and B, along with their corresponding response results. The radar chart accompanying the Q-matrix table illustrates the diagnosed results of the two examinees on the respective dimensions. As depicted in the radar chart, the P-NCD model offers interpretable diagnosis reports, delineating examinees' proficiencies across various dimensions. Then, the dimensional difficulties and the examinees' proficiencies were compared. As shown in Fig. 6, for the 100% stacked bar chart, the varying shades of gray blocks represent the ability to obtain the corresponding score at least, and the dividing values between the blocks correspond to the difficulty of the examinee to obtain more than a specific score. The different colors and markers represent examinee A's and examinee B's proficiencies in each relevant dimension. Fig. 6 shows that when the examinee answered a score, the diagnosed dimensional proficiencies tended to be higher than the dimensional difficulty for this score. For example, item 2 belongs to the dimension of 'Cautious', and the corresponding difficulty is 0.39 (score $\geq$ 1), 0.58 (score $\geq$ 2), 0.78 (score $\geq$ 3). Examinee A's proficiency in 'Cautious' is 0.30, which is lower than the difficulty(score $\geq$ 1), so he scores 0. Examinee B's proficiency on 'Cautious' is 0.48, which is lower than the difficulty (score $\geq$ 2), so he scored 1.

6. Discussion and conclusion

6.1. Discussion

This study presented a novel polytomous cognitive diagnosis framework, which utilizes the cumulative category response function to handle polytomous data and integrates with traditional models for better diagnosis. First, the cumulative category response



(a) Q-matrix and diagnosed result by P-NCDM
(b) Difficulties and diagnosed proficiencies by P-NCDM

Fig. 6. Diagnosed results of two examinees and their relation with dimensional difficulties on FTI. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

function was used to re-encode response data to obtain the input and output vectors for existing cognitive diagnostic models. Next, based on the definitions of the different item parameters, the item parameter one-hot representation vectors were selected to fit the current cognitive diagnostic model. Finally, the cumulative category response function was utilized to combine the results and get a final prediction of the examinees' responses. Extensive experiments conducted on real-world datasets demonstrated that PCDF is not only practical and efficient for cognitive diagnosis, particularly in handling polytomous data, but also yields rich and interpretable diagnostic outcomes.

The similarity in results observed between MIRT and NCDM as CDM adaptor may be attributed to the fact that the interaction layer of NCDM draws inspiration from MIRT models (Wang et al., 2020). Additionally, the datasets utilized in this study exhibited a small amount of missing data and a limited number of dimensions (16PF and FTI have 16 and 4 factors, respectively), which could contribute to the minimal discrepancy between the two base models in the results. The relatively poor performance of the IRT base model is understandable given its treatment of examinees' latent abilities as unidimensional, whereas this research employed multidimensional data. Some findings indicate that the baseline model LS slightly outperforms PCDF. However, since LS involves a direct linear division of estimated probabilities, its item parameter settings align with those of the dichotomous model, thereby rendering LS less interpretable than PCDF.

6.2. Conclusion

6.2.1. Implications

In this study, we have applied the principles of CCRF theory to re-encoding the polytomous data. It is worth noting that in previous studies, researchers have primarily utilized CCRF to develop new models (de Ayala, 1994; Henson et al., 2009; Samejima, 1995) rather than employing it for data re-encoding purposes. We verify that the additivity of the score category response function can be employed to purposefully guide the data re-encoding process, and that such re-encoding does not compromise the interpretability of the parameters associated with the polytomous items.

PCDF represents a straightforward and versatile framework that can seamlessly integrate with a wide range of existing dichotomous CDMs, ensuring both model prediction performance and parameter interpretability. While Likert-type data were employed in this study, PCDF's applicability extends to various question types in hierarchical scoring, including subjective questions encountered in intelligent tutoring systems.

6.2.2. Limitation and future work

There are some limitations in this study. Firstly, the datasets we used have a maximum of only 16 dimensions, which is less than other educational datasets like ASSIST0910.[4] This lower dimensionality may prevent the deep learning-based models from fully demonstrating their effectiveness. Secondly, this study did not use educational datasets to test the effect of PCDF, despite the theory of CCRF and its practical applications suggesting its suitability for educational datasets (Matteucci & Stracqualursi, 2006). This decision was due to the presence of subjective items in the public dataset MATH2015,[5] which contains a large number of graded scores in a small number of subjective items. This setup is not ideal for demonstrating how an increase in the number of graded scores affects the model's estimation of examinees' responses. Researchers interested in this area might consider merging the data

[4] https://sites.google.com/site/assistmentsdata/home/2009-2010-assistment-data.

[5] http://staff.ustc.edu.cn/%7Eqiliuql/data/math2015.rar.

in MATH2015 by category, which would provide more detailed information about item parameters compared to mapping scores onto [0, 1]. Finally, the method proposed in this study is based on the assumptions of graded scores data and is only applicable to it. PCDF cannot handle data types such as nominal data (Darrell Bock, 1972; Muraki, 1992), and future studies may consider designing models specifically for nominal data.

Future research can be conducted on these aspects. First, as the number of score categories increases, the accuracy of the model predictions decreases. Five is the maximum number of score categories considered in this study, future research could be intriguing if it explores a broader spectrum of score categories. Second, Q-matrix information is extracted from item contents (Cheng et al., 2019; Wang et al., 2020; Zhu et al., 2018), considering the item content information enables the model to predict the examinees' responses better. Third, when considering additional factors influencing models, it is important to acknowledge that examinees may exhibit different response biases depending on the context of their responses, such as social desirability and malingering. These biases could potentially impact the accurate assessment of the examinee's ability. Finally, the use of imbalanced data can have an impact on the performance of deep learning-based CDMs, so it is necessary to include it as a consideration in future research.

CRediT authorship contribution statement

Xiaoyu Li: Writing – review & editing, Writing – original draft, Validation, Methodology, Conceptualization. **Shaoyang Guo:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Jin Wu:** Writing – review & editing, Validation, Methodology. **Chanjin Zheng:** Writing – review & editing, Supervision, Methodology.

Data availability

Data will be made available on request.

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